Fish Biology

Introduction to Fish Biology

- About 25,000 species of fish have been described this is the most numerous group of vertebrates
- Approximately 45%-50% of all vertebrate species are fish dominant life form in almost all aquatic systems.

Common characteristic of fishes

- cranium backbone
- live in water
- possess gills throughout life
- they either have or are descended from ancestors that had y scales or armor derived from dermal tissue

Fish shapes

- Cold-blooded aquatic animals with backbones, gills, and fins
- Most are torpedo-shaped (fusiform) for efficient travel through water, but they may be flattened and rounded (flounder), and vertical and angular (sea horses)

Fish Size

- Fish range in size.
- the pygmy goby, which reaches only 12mm long and 1.5g in weight sexually mature at 6mm.
- The whale shark grows to 18m and weighs over 20 tons.
- The tiny coral reef-dwelling fish called the pygmy goby has taken the record as the shortest-lived vertebrate. The pygmy goby lives an average of 59 days, pipping the previous record holder, an African fish which lives for just over two-and-a-half months.

Fish Distribution

- Found throughout the world
- Altitudes of more than 5000m to depths of about 10km
- Some inhabit hot springs, such as Cyprinodon, where water may reach 45C
- Others are found in the Antarctic seas, such as Chaenocephalus, where it may be less than 0C
- 107 species are distributed worldwide in tropical and subtropical waters, but many species have limited ranges

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- There are 28 000 or more fish species
  - 60 live in marine waters
  - 40 are found in fresh water
- Most of the worlds fish are continental – either in freshwater on land, or in the sea close to the coast.
- Coasts are good for fish as they are rich in nutrients such as nitrates and phosphates from rivers, upwelling, aeration from the surf and tide, and the sunlight.

Fish Anatomy

Three classes

- Agnatha (jawless fishes) that consists of hagfish and lampreys
- Chondrichthyes (cartilaginous skeleton) that consists of sharks and rays
- Osteichthyes (bony-skeleton) all other fish

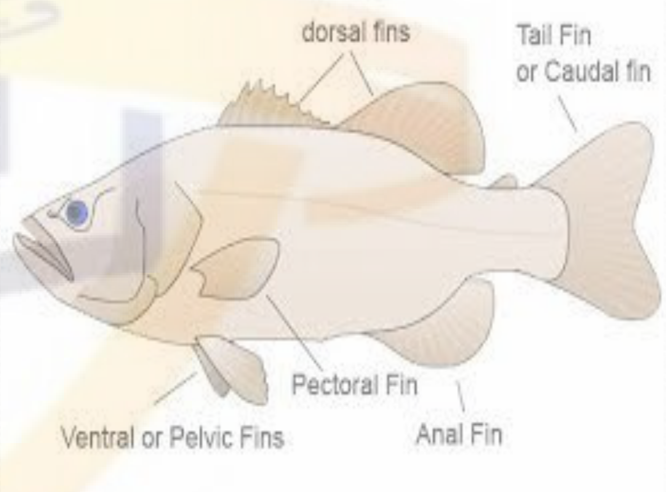
Skeletons

Skeletons of the three groups are very different

- Hagfish and lampreys have a notochord, a rod-like structure composed of unique tissue
- Sharks and rays have a notochord that is surrounded and constricted by the vertebrae, to form a backbone. The remainder of their skeleton is composed of cartilage, not bone (but it can be hardened by calcareous salts)
- Primitive bony fishes have vertebrae that are mostly cartilaginous, but advanced fish have bony vertebrae united to form a backbone, with no notochord

Fins

- Fins are either median or paired
- Median fins are along the centerline of the body, at the top, bottom, and the end
- Dorsal fin (top) may be one or several fins (one behind the other), and may include a fleshy fin (adipose) near the tail
- Bottom (anal) fin is located on the belly behind the anus
- End fin is called the tail, or caudal fin



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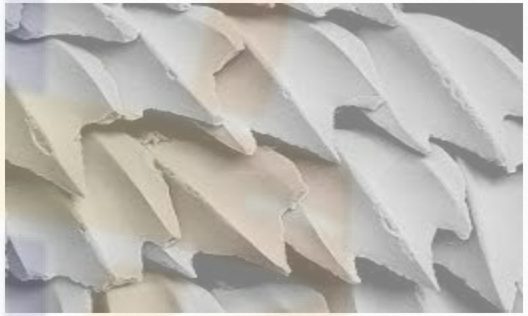
- Pectoral fins are at the front of the body, behind the gill openings and usually provide maneuverability
- Pelvic fins (ventral fins) are along the bottom of the body but can vary in their placement (can be in the middle of the belly, below the pectorals, or in front of the pectorals.) Can provide maneuverability

Scales

- Scales are colorless
- Coloring of the fish comes from the structures beneath or associated with the scales
- Not all species of fish have scales, or they can be so tiny it would appear that they do not have any
- Scales may only be present on small areas of the body
- Arranged in imbricate (overlap like shingles) or mosaic (fitting closely together or barely separated)

Four Basic Scale Types

- **Placoid scales** (dermal denticles) are on sharks and rays, and
- are tooth-like in structure (enlarged ones are actual teeth in sharks)
- Scales do not increase in size, so new scales must be added as the shark grows
- **Cosmoid scales**
- Coelacanth have them
- Also in lungfishes (only single-layered)
- Four-layered bony scale



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- **Leptoid scales**
- Derived from ganoid scales by the loss of the ganoin layer
- Single layer of bone
- Found on the higher bony fishes and occur in two forms cycloid (circular) and ctenoid (toothed)

Fish Circulation

- Blood transports oxygen, nutrients, and wastes
- Single circuit heart-gills-body-heart
- Fish heart is two-chambered, with an upper atrium and lower ventricle
- Other organisms may have three or four-chambered hearts

Fish Respiration

- Fish must extract oxygen from the water and transfer it to their bloodstream
- This is done by gills, lungs, specialized chambers, or skin
- More difficult than extracting oxygen from air
- Some fish can extract as much as 80% of the oxygen in the water passing over the gills whereas humans can only extract about 25% of the oxygen from the air taken into lungs

Gills are efficient because

- Large surface area gills have 10 to 60 times more than body surface area
- Short distance for oxygen to diffuse blood in the gills is close to the water
- Countercurrent circulation blood flows forwards while the water flows the other so that there is always less oxygen in the blood than in the water
- Water flows one direction only over the gills